

EVOLUTION

The genetic basis of a recent transition to live-bearing in marine snails

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Key innovations are fundamental to biological diversification, but their genetic basis is poorly understood. A recent transition from egg-laying to live-bearing in marine snails (*Littorina* spp.) provides the opportunity to study the genetic architecture of an innovation that has evolved repeatedly across animals. Individuals do not cluster by reproductive mode in a genome-wide phylogeny, but local genealogical analysis revealed numerous small genomic regions where all live-bearers carry the same core haplotype. Candidate regions show evidence for live-bearer-specific positive selection and are enriched for genes that are differentially expressed between egg-laying and live-bearing reproductive systems. Ages of selective sweeps suggest that live-bearer-specific alleles accumulated over more than 200,000 generations. Our results suggest that new functions evolve through the recruitment of many alleles rather than in a single evolutionary step.

Evolution is a gradual process but occasionally results in sudden changes in form and function that allow organisms to exploit new ecological opportunities (1, 2). These game-changing traits, including flight, vision, and the bearing of live offspring, are known as “key innovations” (2–5). Key innovations are all around us and have catalyzed the diversification of many groups (1). Despite their significance, we know surprisingly little about the origins and genetic basis of innovations (1). This is because most of them originated deep in the past, making it difficult to disentangle causal loci from the countless other genetic changes that accumulated up to the present.

A recent transition in female reproductive mode offers a rare opportunity to study the genetic basis of an innovation that has evolved many times across the animal kingdom (6). We focus on a clade of intertidal gastropods (genus *Littorina*) in which the ancestral state is to lay a large egg mass, but one species gives birth to live young (Fig. 1A and fig. S1) (7, 8). Egg-layers have a gland that embeds fertilized eggs into a protective jelly. In the live-bearer, *L. saxatilis*, this structure has evolved into a

brood pouch where embryos develop inside the mother. Live-bearing is the only taxonomic character that is diagnostic of *L. saxatilis* (8), because no other known trait differs consistently between the live-bearing and egg-laying individuals (8). In fact, sympatric populations are so similar that the difference in mode was long thought to reflect within-species polymorphism (8, 9), and molecular markers are needed to identify males and juveniles in locations where live-bearing and egg-laying species coexist (10).

Live-bearing is thought to be an adaptation that allows snails to reproduce in areas where eggs would be exposed to harsh conditions (8). This is reflected in the much broader ecological and geographic distribution of *L. saxatilis* compared with the two closely related egg-laying species, *L. arcana* and *L. compressa* (8) (Fig. 1, B and C, and fig. S2). Egg-laying and live-bearing species have adapted in parallel to contrasting environments across the intertidal zone (8, 11), largely decoupling reproductive mode from other axes of phenotypic divergence (Fig. 1B). There is also evidence for rare gene flow between egg-layers and live-bearers (10). These features provide an opportunity to identify and study the genetic changes underlying the live-bearing innovation.

Live-bearing snails do not form a monophyletic group

We used whole-genome sequences from 108 individuals to test the existing hypothesis of a single origin of live-bearing, which was inferred by parsimony analysis in earlier phylogenetic studies (Fig. 1D, figs. S3 and S4, and tables S1 and S2) (7, 8). Rather than forming a single clade, live-bearers formed two separate, well-supported clades in a genome-wide phylogenetic tree (Fig. 1E). One clade contains all *L. saxatilis*

from Iberia (hereafter “Iberian *saxatilis*”) and another includes all other *L. saxatilis* (hereafter “northern *saxatilis*”) that were sister to a clade of egg-laying *L. arcana* (the same pattern was observed in an ML tree, phylogenetic network, and principal components analyses; figs. S5 and S6).

The discordance between evolutionary relationships and reproductive mode has several possible explanations. One interpretation of the genome-wide tree is that there has been more than one transition between egg-laying and live-bearing. However, because the tree represents the aggregate signal from all loci in the genome, it does not necessarily reflect the evolutionary history of these groups or of any single locus or trait (12). This means that it is also possible that live-bearing may have evolved once and causal alleles became associated with two different lineages through the interaction of gene flow and selection (12). If this were the case, then we would expect genealogies for loci that cause live-bearing to be strongly discordant from the genome-wide tree, with samples grouping by reproductive mode.

Topology weighting reveals rampant genealogical discordance and loci associated with reproductive mode

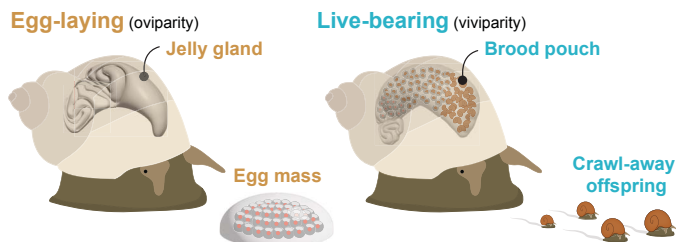
With the above expectation in mind, we used topology weighting (Fig. 2A) to identify genomic regions associated with reproductive mode. For each genomic window, topology weighting calculates the degree of monophyly toward three possible taxon subtrees (Fig. 2, B and C, and fig. S7): (i) the background topology (Tb) observed in our genome-wide analysis; (ii) the reproduction topology (Tr), in which samples cluster by reproductive mode; and (iii) the control topology (Tc), which is of no specific interest but provides a control for distinguishing incomplete lineage sorting from other processes that cause genealogical discordance (e.g., gene flow). We used non-overlapping 100–single nucleotide polymorphism (SNP) windows (mean size, 5.8 kb; fig. S8) and calculated topology weights (13) for each window by sampling 10,000 subtrees (Fig. 2A).

We analyzed the joint distribution of topology weights in a ternary framework. This approach exploits the geometric properties of the ternary plot, allowing us to visualize and quantify various properties of the genome-wide distribution of discordance (Fig. 2A). We used simulations to illustrate how different factors, including the timing of population splits and gene flow between nonsister lineages, shape the ternary distribution of topology weights for large numbers of loci (Fig. 2B, supplementary text, figs. S9 to S19, and tables S3 and S4).

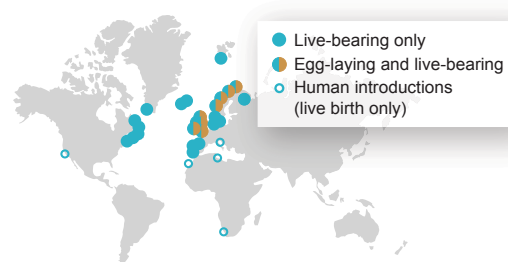
We expected the empirical distribution of weights to be biased toward Tb because this

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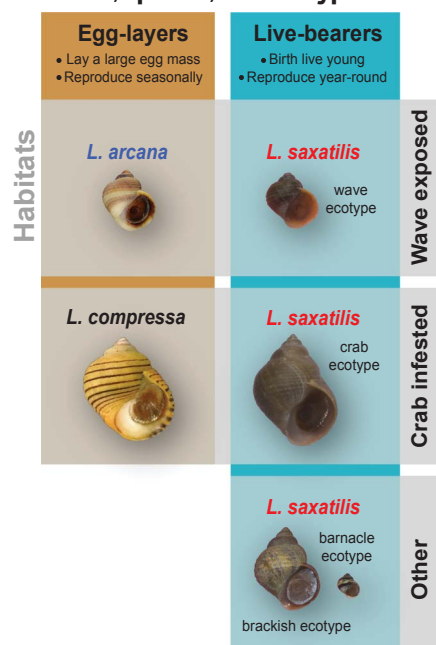
A Reproductive modes



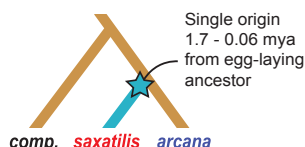
C Distributions of egg-laying and live-bearing



B Modes, species, and ecotypes



D Hypothesis for the origin of live-bearing



E Whole-genome NJ phylogeny

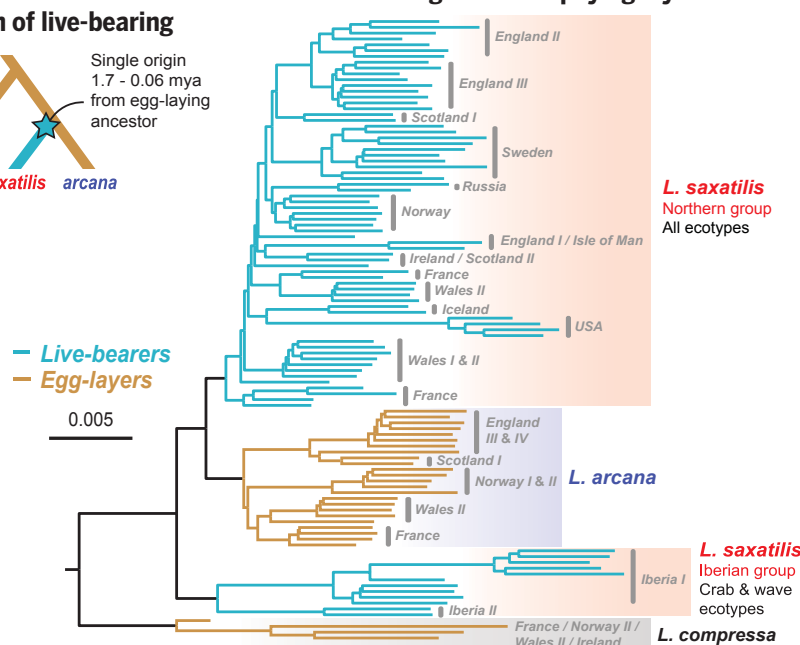


Fig. 1. Variation in reproductive mode in *Littorina*. (A) Anatomical differences between modes. (B) Egg-layers reproduce during a limited breeding season, whereas live-bearers release offspring year-round. The two egg-layers share their habitats with ecotypes of the live-bearer *L. saxatilis*. (C) Approximate distributions of the modes

highlighting the broader distribution of live-bearing. (D) Existing hypothesis for the origin of live-bearing inferred by past phylogenetic studies (7, 8). (E) Neighbor-joining phylogenetic tree based on whole-genome sequences (108 individuals and 18.5 million variable sites). All nodes have 100% bootstrap support.

was the topology observed in the genome-wide analysis. However, the observed bias was only slight ($T_b = 0.380$, $T_c = 0.310$, and $T_r = 0.308$), with just 62 of ~155,000 genomic regions perfectly fitting T_b (i.e., $T_b = 1$) (Fig. 2C). Instead, the bulk of the distribution fell close to the center of the triangle. This indicates that sequence variation is broadly shared between groups, a pattern that could be caused by extensive incomplete lineage sorting due to rapid diversification relative to the effective population size, widespread gene flow during divergence, or both (14, 15). Thus, although well supported statistically, the genome-wide tree is a very poor predictor of evolutionary relationships at any given genomic region.

We found substantial left-right asymmetry in the distribution of topology weights (Fig. 2D). Such a bias is not expected to arise from incomplete lineage sorting because there is an equal chance that a given gene tree will more closely resemble either alternative topol-

ogy (Fig. 2B and supplementary materials) (14). We detected asymmetry using a new statistic, D_{LR} (Fig. 2D and fig. S19). A genome-wide test performed by calculating D_{LR} between the two halves of the triangle, revealed a 3.4% excess of windows shifted toward the control topology ($D_{LR} = 0.034$, permutation test $P = 1 \times 10^{-5}$). D_{LR} calculated between analogous left and right subtriangles revealed that this asymmetry was driven by an excess of trees with a small bias toward T_c (Fig. 2D and table S5). Further exploration showed that this bias was caused by 10 previously identified chromosomal inversions (16), none of which was associated with reproductive mode (figs. S20 to S23, table S6, and supplementary materials). For each inversion, one arrangement was more common in Spanish *L. saxatilis* and *L. arcana*, and the other was more common in *L. compressa* and northern *L. saxatilis*. When the chromosomal inversions were removed, we found no significant left-right asymmetry at the genome-

wide level (D_{LR} for regions outside inversions = -0.007 , $P = 0.074$; fig. S20).

Much stronger asymmetry was observed between the far left and right subtriangles, corresponding to windows that more strongly fit one of the alternative topologies (Fig. 2D and fig. S24). However, the asymmetry was in the opposite direction from the genome-wide pattern, with a large excess of windows strongly biased toward the reproduction tree compared with the control tree ($T_r > 0.7 = 1151$ windows versus 461 for T_c ; $D_{LR} = -0.43$, $P = 1 \times 10^{-5}$). A total of 88 windows perfectly fit the reproduction topology (i.e., $T_r = 1$; table S7) compared with no windows that perfectly fit the control topology ($D_{LR} = 1.00$, $P = 1 \times 10^{-5}$; Fig. 2E and fig. S25).

Evidence for live-bearer-specific positive selection

Although neutral gene flow can generate strong asymmetry under some circumstances, we

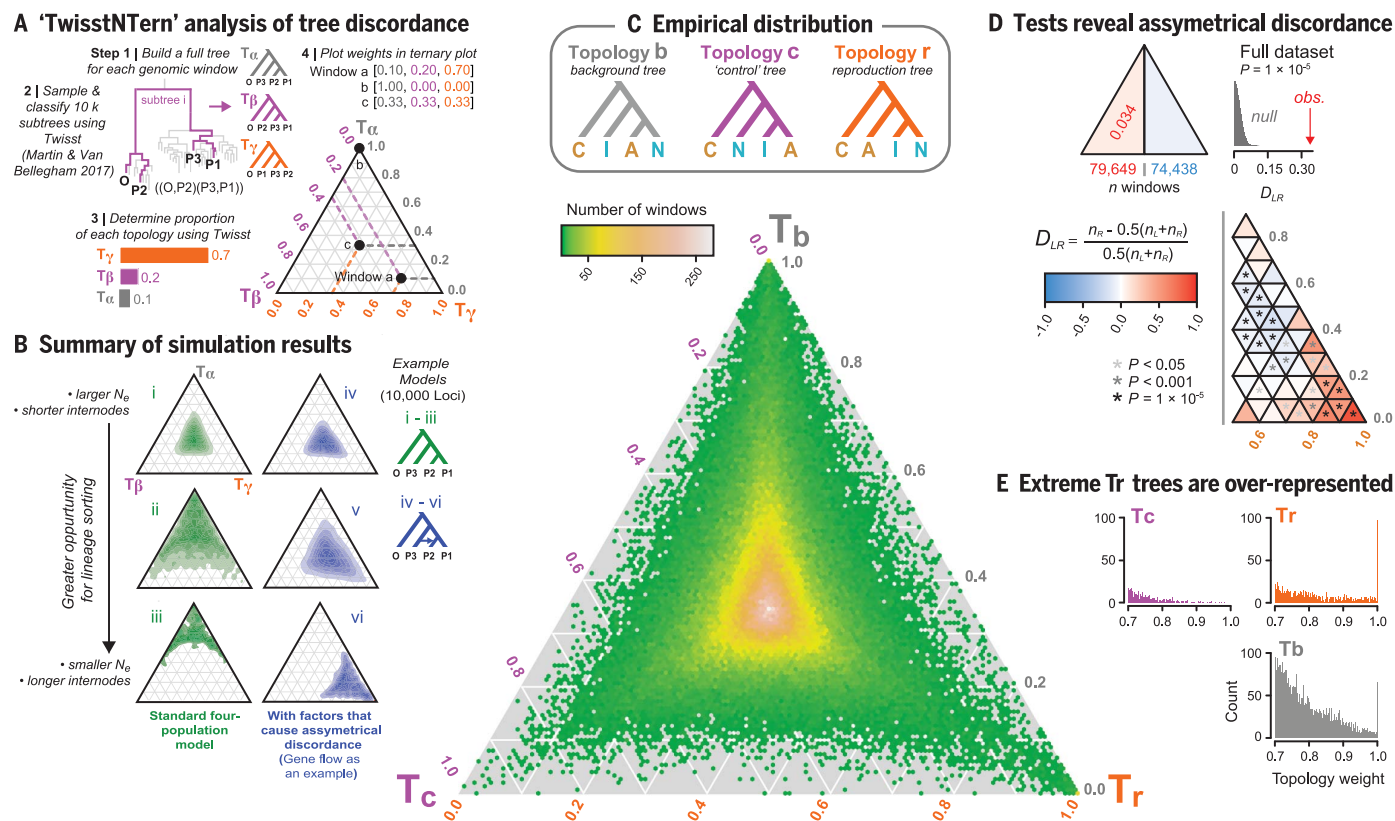


Fig. 2. Topology weighting reveals genomic regions associated with reproductive mode. (A) For each genomic window, we inferred a tree for all haplotypes and then classified 10,000 “subtrees” by randomly picking one haplotype per group. Topology weights are the proportions of each topology among all subtrees. Windows were plotted in a ternary plot based on the weights. (B) Simulated distributions of weights. A greater opportunity for lineage sorting (i to iii) biases the distribution toward the topology that matches the demographic history. Incomplete lineage sorting yields genealogies that are a better fit to one of the discordant trees, but the distribution is always

symmetrical between the left and right half-triangles. Additional factors, including gene flow, create a bias toward one discordant genealogy (iv to vi). (C) Possible topologies and the empirical distribution of weights for the 154,971 100-SNP windows (“C,” *compressa*; “A,” *arcana*; “I” Iberian *saxatilis*; “N,” northern *saxatilis*). Hexagonal bins are colored by window count. (D) Counts of windows in the left and right half-triangles, with asymmetry quantified using D_{LR} . Further division into subtriangles reveals left-right asymmetry throughout the distribution. Asterisks indicate significant asymmetry between corresponding left- and right-sided subtriangles. (E) Distributions of weights >0.7 .

are unable to explain the observed T_r bias without invoking natural selection (see the supplementary materials and table S8). We found strong additional evidence for live-bearer-specific positive selection in regions associated with reproductive mode. First, window-based estimates of nucleotide diversity (π) in live-bearers decreased substantially with increasing T_r weight (Fig. 3A), but we found no such relationship in egg-layers. Eighty-four (95%) of the 88 perfectly associated regions showed reduced π in live-bearers (mean $\pi_{\text{live-bearer}} = 0.0029$ versus $\pi_{\text{egg-layer}} = 0.0065$; paired Wilcoxon test, $P = 1.313 \times 10^{-15}$; Fig. 3A and fig. S26). These results are consistent with selection having purged diversity from haplotypes associated with live-bearing (17). Although this result could in principle result from a live-bearer-specific demographic bottleneck, we can rule this out because live-bearers and egg-layers have similar levels of genome-wide diversity (mean $\pi_{\text{live-bearer}} = 0.0065$ versus $\pi_{\text{egg-layer}} = 0.0062$; Fig. 3A and fig. S27). Further, relationships between π and the other weights

(T_b and T_c) were weak and similar for both groups, confirming that reduced π in live-bearers is specific to T_r rather than being a general feature of windows with extreme weights (fig. S28). The site-frequency spectra (SFS) and sample-size-corrected estimates of private alleles for perfectly associated regions provide further evidence for selection (Fig. 3, B to D; figs. S29 to S31; and tables S9 to S11): the live-bearer SFS was strongly skewed toward rare variants [Tajima’s $D = -1.89$, 95% confidence interval (CI) = -1.77 to -2.01 ; fig. S29], the majority of which (80%) were private to live-bearers. Both results are expected during the phase when diversity is recovered by mutation after a selective sweep (18). By contrast, the SFS for egg-layers was much closer to the neutral expectation (Tajima’s $D = -0.24$, 95% CI = -0.037 to -0.437), with polymorphic sites being 2.14 times more abundant in egg-layers after accounting for the difference in sample size.

We next characterized footprints of selection within contigs to estimate the number and size of candidate regions more accurately

(Fig. 3E). The 88 perfectly associated windows mapped to 50 contigs in our genome assembly (mean 1.7 ± 1.5 windows per contig; table S9). Associated regions were narrow, mostly spanning <20 kb (mean 12 ± 14.4 kb). Sliding-window analysis of each contig generally revealed clear peaks of allele frequency differentiation (F_{ST}) and sequence divergence (d_{xy}) between the egg-layers and live-bearers, as well as valleys of π in live-bearers (Fig. 3E). We also inferred ancestral recombination graphs (ARGs) for selected contigs to refine candidate regions (Fig. 3E). Unlike trees inferred from genomic windows with arbitrary start and end positions, each tree in an ARG corresponds to an inferred nonrecombining segment of the genome (19). Thus, by applying topology weighting to the sequence of marginal trees, we were able to identify more precisely the segment of genome retained by all live-bearing samples after the selective sweep. In both cases, the core live-bearing haplotype spanned <2 kb. Live-bearers showed much shallower coalescence in these regions than egg-layers, as expected after a sweep (Fig. 3E).

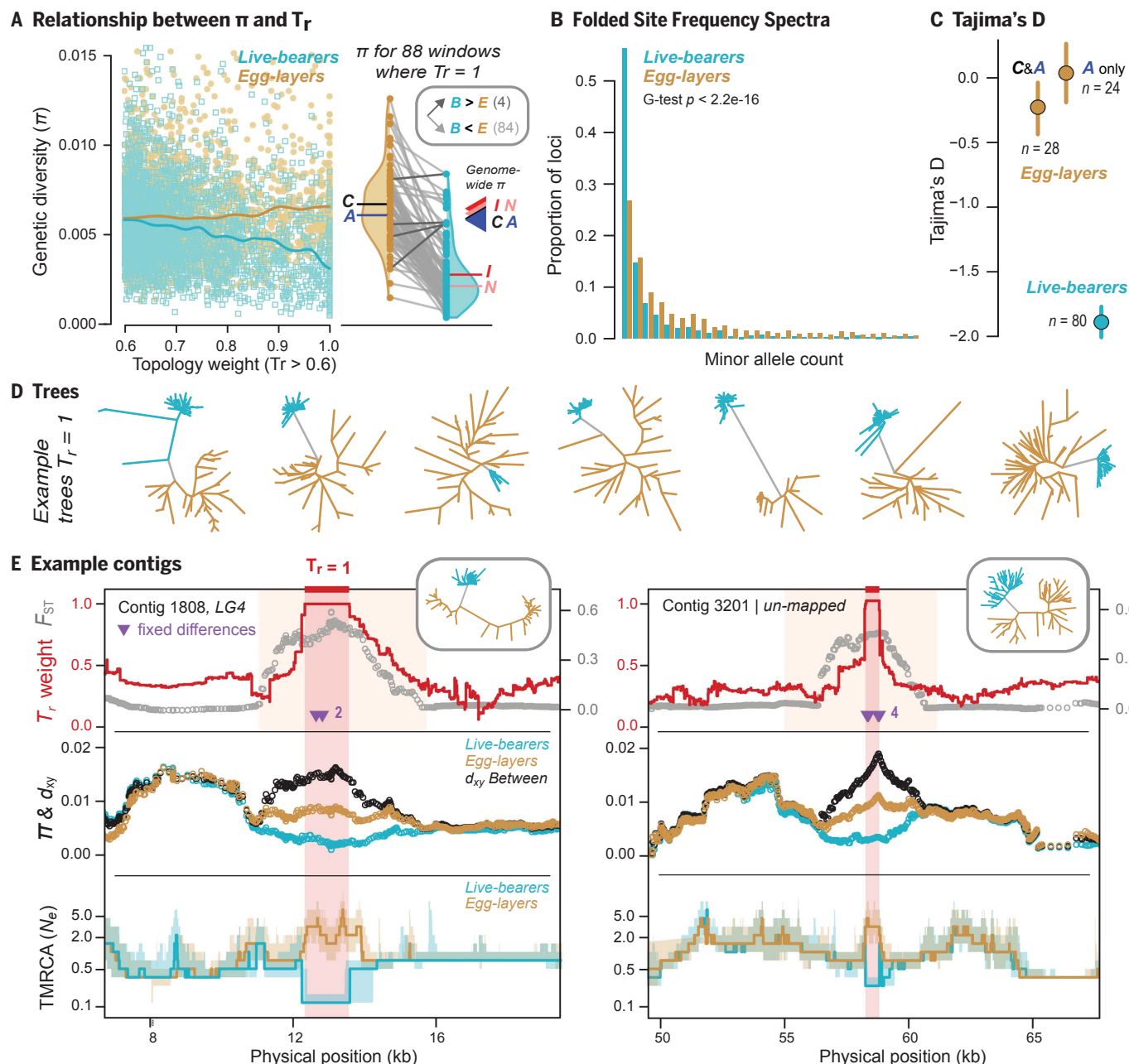


Fig. 3. Evidence for positive selection on haplotypes associated with live birth. (A) Relationship between π and T_r for both reproductive modes. Triangles on right indicate genome-wide π . Violin plots indicate distributions of π for windows where $T_r = 1$. (B) Folded SFS for each mode in perfectly associated regions projected at the same sample size for comparison. (C) Estimates of Tajima's D with 95% CIs for perfectly associated regions. (D) Examples of trees for windows where $T_r = 1$. (E) Variation across two example contigs that contain a window where $T_r = 1$ (span of the orange box). The tree associated with

each region is shown. Top panel: F_{ST} between egg-layers and live-bearers in 3-kb sliding windows (30-bp steps). T_r shows the results of topology weighting applied to marginal trees obtained from inferred ARGs. Purple arrows show fixed differences between modes. Middle panel: π and d_{xy} in sliding windows. Bottom panel shows traces of time to the most recent common ancestor (TMRCA) obtained from ARGs. Bold lines are median estimates. Envelopes are 95% CIs. The red box shows the inferred length of the core haplotype block associated with live birth.

Approximate estimates of the timing of each selective sweep at the mode-associated loci, determined from the accumulation of private mutations ($T = \pi_{\text{private}}/2\mu$), span a broad range from ~20,000 to 200,000 generations before present, with a median of 70,000 generations before present (fig. S32). Assuming two generations per year, this equates to 100,000

to 10,000 years before present, with a median time of 35,000 years.

Mode-associated regions are widespread and enriched for genes that are differentially expressed between reproductive systems

The assignment of contigs to a genetic map revealed that reproductive-mode-associated

windows are widespread across the genome rather than colocalizing to one or a few genomic regions (Fig. 4A). As expected for a polygenic trait, the number of mode-associated windows on each linkage group (LG) was strongly predicted by LG size ($T_r > 0.7$, $r = 0.79$, $P < 0.0001$; $T_r > 0.9$, $r = 0.71$, $P < 0.005$). Associated regions were also widespread within linkage



Fig. 4. Candidate regions are widespread across the genome and enriched for DEGs between reproductive systems. (A) The number of high T_r windows ($T_r > 0.7$) assigned to each of the 17 *L. saxatilis* LGs. Dots show the expected number of windows given the total assigned to each LG. (B) Distribution of high- T_r windows across LGs. (C) Clustering of tissues by expression and the number of DEGs in each expression class. (D) Clustering of reproductive tissues based on patterns of expression. (E) The proportion of genes in each differential expression class after binning genes according to the T_r weight.

groups, in some cases with strong associations near opposite ends of the same LG (Fig. 4B).

Candidate regions also showed strong enrichment of genes that are differentially expressed between female live-bearing and egg-laying reproductive tissues. To identify differentially expressed genes (DEGs), we collected reproductively mature female *L. arcana* and northern *L. saxatilis* at peak breeding season from a single location (to control for environmental effects) where sympatric egg-layers and live-bearers are morphologically cryptic aside from their reproductive anatomies. We first compared transcriptomes from pools of reproductive systems (brood pouch versus jelly gland) and foot tissue paired from the same individuals. Clustering analysis based on patterns of gene expression (13,492 genes) revealed that pools of reproductive tissue grouped by system type, but egg-laying and live-bearing species did not group based on expression in foot tissue. Differential expression analysis revealed

1044 DEGs and showed much higher rates of tissue-specific differential expression between reproductive systems (Fig. 4C and fig. S33). To increase power to detect DEGs between the reproductive systems, we sequenced additional pools of reproductive tissue (Fig. 4D and fig. S34). This analysis detected 1450 DEGs, 66.1% (858) of which showed higher expression in the brood pouch of live-bearers. To test for the enrichment of DEGs in regions associated with reproductive mode, we binned each DEG according to the T_r score of its associated genomic region (Fig. 4E, fig. S35, and table S12). We found that the proportion of reproductive mode DEGs strongly increased with increasing T_r weight (Spearman's $\rho = 0.903$, $P = 9 \times 10^{-4}$) (table S13). No correlation was observed between T_r weight and foot-tissue only DEGs (Spearman's $\rho = -0.410$, $P = 0.217$) (table S13).

Gene ontology (GO) analysis and functional annotation suggested that the transition to

live birth involved genes with diverse functions. Separate GO analyses conducted on a sequence-based gene set (574 genes in regions where $T_r > 0.7$) and expression-based gene set (1450 reproductive mode DEGs) yielded 37 enriched GO terms, including transmembrane transport, calcium-ion binding, and ion channel activity (fig. S36). We examined the putative functions of the 27 genes found in both sets in more detail (table S14). These included genes putatively associated with antibacterial activity (lectin L6-like protein; higher expression in the brood pouch), the synthesis of mucin-type oligosaccharides (GALNT10-like; higher expression in the brood pouch), the formation of structural tissue (IFB-like and CMP-like, both higher expression in the brood pouch), and two secretory genes involved in egg mass production in another marine snail (both with lower expression in the brood pouch).

Conclusions

Our analyses show that live-bearing in *Littorina* is associated with selection on many loci, as was the case in the only comparable analysis in *Zootoca* lizards (20). Although our genome-wide analysis suggested two independent origins of live-bearing, the high sequence similarity of live-bearer-specific alleles indicates that they had a single origin. Given the number of associated loci, the history of their origin and spread may be highly complex, potentially varying among loci. One possible scenario is that all live-bearer-specific alleles originated in one location (e.g., in the ancestor to the Iberian clade), after which the range of live-bearers expanded until they encountered egg-layers. Hybridization may have then allowed beneficial live-bearing alleles to introgress onto the local egg-laying genetic background, or it may have eroded genome-wide differentiation between egg-layers and live-bearers, whereas selection maintained alternative sets of alleles at mode-associated loci. It is also possible that alleles arose in numerous locations, and that associations built up between them at different times, perhaps as live-bearing spread. Regardless of the precise history, which we cannot resolve at present, the interaction between gene flow and selection has allowed us to identify loci associated with reproductive mode.

Live-bearing is the only known trait that consistently distinguishes *L. saxatilis* from the egg-laying species, making associated loci good candidates for causing the difference in reproductive mode. We found supporting evidence in our expression analysis, because associated regions were strongly enriched for DEGs between the reproductive systems. This suggests that selection has acted on differences in gene expression, driving the evolution of live-bearing, including the development of the brood pouch. Because reproductive mode is a complex trait,

associated loci may also underpin a diverse range of biological functions, including differences in the synchronization of egg production (8), differences in embryo retention times (21), and variations in immune function and metabolism (21). It is, however, important to emphasize that some loci may not be causally associated with reproductive mode and may instead underlie other less conspicuous traits that are functionally linked or associated with the live-bearing or egg-laying lifestyle.

Polymorphic inversions often underpin local adaptations (22, 23) and are thought to maintain beneficial sets of alleles by suppressing recombination (24). This makes a role in the evolution of a key innovation seem likely, but we found that known chromosomal inversions in *Littorina* are not associated with the difference in reproductive mode. Many large inversions are shared among these *Littorina* species (16), and there is evidence that they play a key role in ecotype formation and reproductive isolation (25–27). For example, in *L. saxatilis*, repeated adaptation to contrasting crab-infested and wave-swept environments usually involves 8 to 12 major inversions, with the alternative arrangements contributing to differences in morphology and behavior (26, 27). However, egg-laying species coexist with *L. saxatilis* and likely use the same inversions to adapt to local selection pressures (16). Thus, the independence of genetic architectures for ecotype formation and reproductive mode may be a major factor permitting the local coexistence of egg-laying and live-bearing species.

Our estimates of the timing of sweeps suggest that alleles associated with live-bearing were recruited gradually over the past 200,000 generations (~100,000 years). This finding is relevant to long-standing debate about the genetic basis of evolutionary novelty. Because

key innovations are not visible to selection before they arise, models of saltational evolution invoke large-effect macromutations to explain their evolution (28). We do not know which mutation caused the threshold from egg-laying to live-bearing to be crossed. Some potentiating mutations may have preceded live-bearing but were critical to its origin, and others may have refined live-bearing after it arose. Nevertheless, our results suggest that new functions evolved gradually through the recruitment of alleles at many loci rather than arising in a single evolutionary step (20, 29, 30).

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SUPPLEMENTARY MATERIALS

science.org/doi/10.1126/science.adi2982

Materials and Methods

Supplementary Text

Figs. S1 to S36

Tables S1 to S14

References (32–79)

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The genetic basis of a recent transition to live-bearing in marine snails

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Editor's summary

Traits are constantly changing in populations, but the emergence of a trait that provides a new function is a rarer occurrence. Two studies now examine how such traits arise. Stankowski *et al.* identified the multiple genome regions associated with the transition from egg-laying to live-bearing reproduction in a clade of snails, demonstrating a polygenic basis of live-bearing reproduction with alleles that accumulated in response to selection over time. Chomicki *et al.* found that the spontaneous occurrence of three variable traits created a novel mechanism for trapping insects in pitcher plants, which arose separately in two different species. Together, these studies provide insight into the emergence of complex, transformative traits (see the Perspective by Elmer). —BEL and CNS

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